Brain Interacts

WHITEPAPER

The Sunk Cost Trap

Why enterprises fail to kill doomed projects, and how to fix it.

The Behavioral Enterprise Series

Brain Interacts

Written for

CEOs · CFOs · Heads of Strategy and Portfolio · PMO Leaders · Investment Committees · Board Members

AT A GLANCE

Four ideas in one page

The problem

Large organizations keep funding initiatives long after the evidence says stop. The money is not the only loss; the people and capital tied up cannot be used on better bets.

The insight

Stopping a project means admitting a loss, and people are strongly biased against realizing losses. The bias is predictable, so the fix is structural, not a plea for more courage.

The method

The Behavioral Stage-Gate: six design rules that pre-commit to kill criteria, fund projects in evidence-based tranches, separate decision-makers from sponsors and make stopping a normal, rewarded decision.

The payoff

Earlier terminations, capital redeployed to higher-return work and fewer catastrophic overruns.

THE SENTENCE TO LISTEN FOR

"We have come too far to stop now."

THE SCENE

The project everyone knew was failing

Every large company has one. A transformation program enters its third year. The deadline has moved twice and the budget once. The business case has been rewritten so that the benefits line still clears the hurdle rate.

In steering committee meetings, the most common sentence is some version of "we have come too far to stop now." Nobody believes the original plan. Nobody proposes to end it either. A new sponsor, who might have walked away, inherits the program and the commitment that comes with it.

The data suggests this is not a rare event. Flyvbjerg and Budzier (2011) analyzed 1,471 IT projects and found an average cost overrun of 27%. That average hides a fat tail: one in six projects was what they call a "black swan," with a cost overrun of 200% on average and a schedule overrun of almost 70%.

1,471	27%	1 in 6	200%
IT projects analyzed	average cost overrun	projects a "black swan"	average cost overrun for black swans

Source: Flyvbjerg & Budzier (2011), Harvard Business Review.

They cite an example in which a project of roughly $5 million at Levi Strauss was followed by a $192.5 million charge against earnings and the departure of the CIO. The lesson is not that technology projects are uniquely risky. It is that the worst outcomes are far worse than averages suggest, and they are the ones most likely to be funded past the point where a clear-eyed review would have stopped them.

THE HIDDEN COST

The money is only part of the damage

The financial cost of a late stop is only part of the damage. Every month a failing project consumes its team, its budget and executive attention is a month those resources are not available for something with a better chance.

Opportunity cost is invisible in the budget, which is one reason it rarely features in the discussion about whether to continue.

TEAM

People locked into work that will not pay off

BUDGET

Capital that could fund a better bet

ATTENTION

Executive time spent defending the past

> *The persistence of doomed projects is less a failure of intelligence than a predictable result of how humans and organizations handle losses.*

This paper also shows that the remedy is practical: design the decision process so that stopping is a routine, expected outcome of the evidence rather than an act of personal defeat.

THE SCIENCE

Why stopping feels like losing

Sunk cost

Arkes & Blumer, 1985

The sunk cost fallacy

The tendency to continue an endeavor because of resources already invested, even though those resources cannot be recovered and should not affect a forward-looking decision.

In a classic field study, Arkes and Blumer sold season theater tickets at full price and at two discounts. Buyers who paid full price attended more plays in the following months than those who paid less, even though the tickets and plays were identical.

In questionnaire studies, people who had already invested in a project also rated its chance of success higher than people who had not, and the effect was not reduced by prior economics training.

~2×

Kahneman & Tversky, 1979 · Tversky & Kahneman, 1992

Loss aversion and prospect theory

People evaluate outcomes relative to a reference point and weigh losses more heavily than equivalent gains; later estimates put the weighting at roughly twice as heavy.

Stopping a project converts an unrealized, uncertain loss into a realized, certain one. Continuing keeps alive a chance of recovery, so a rational-looking gamble on "one more quarter" feels better than accepting the loss.

WHY STOPPING FEELS LIKE LOSING, CONTINUED

Escalation

Staw, 1976 · Staw & Hoang, 1995 · Keil, Mann & Rai, 2000

Escalation of commitment

A social and personal layer. In a series of experiments, Staw found that people who were personally responsible for an earlier decision committed more resources to a failing course of action than people who were not, consistent with a need to justify past choices.

Staw and Hoang found the same pattern in a real setting: professional basketball teams gave more playing time and longer careers to players drafted higher, even after accounting for the players' performance.

In software and IT, Keil, Mann and Rai found that project, psychological, social and structural factors combined to produce escalation. Sponsors face reputational stakes; teams lose their jobs when the project ends; and organizations rarely reward people for stopping things.

Inside view

Kahneman & Lovallo, 1993 · Flyvbjerg, 2006

Optimism and the inside view

Optimism feeds the cycle from the start. Planners take an "inside view," focusing on the specifics of their own project, and underestimate the base rate of delay and overrun for similar projects.

The remedy is an "outside view," or reference class forecasting, which anchors estimates on the actual distribution of outcomes for comparable projects.

HONEST LIMITS

The picture is not as tidy as popular accounts suggest. The theater study was small, and later laboratory work found a sunk cost effect that was real but smaller and more erratic than expected (Friedman et al., 2007).

Not all continued investment is irrational: when a project holds real option value, such as learning that can be used elsewhere, or when stopping carries large exit costs, continuing can be the better decision. And loss aversion itself is debated in the research literature.

The practical conclusion is not that every long project should be cancelled.

Decisions should be made on forward-looking evidence, with a process that does not depend on the sponsor's willingness to deliver bad news about their own work.

BACKWARD-LOOKING

Resources already invested

FORWARD-LOOKING

Evidence about what comes next

DIAGNOSIS

How the trap shows up

SYMPTOM IN OPERATIONS	WHAT YOU SEE	LIKELY BEHAVIORAL DRIVER
Serial deadline extensions	The same milestone is re-baselined repeatedly	Loss aversion; hope of recovery
"Too far to stop" language	Past spend is cited as a reason to continue	Sunk cost fallacy
No kill criteria at the start	Nobody can say what evidence would end the project	No pre-commitment; ambiguity protects sponsors
Sponsor-run reviews	The person who proposed the project judges its progress	Escalation of commitment; self-justification
Green status, red reality	Reports stay positive until a crisis	Fear of delivering bad news (see The ROI of Psychological Safety)
Business case drift	Benefits are rewritten to match actual spend	Motivated reasoning
Starved new ideas	Promising proposals wait while failing ones consume the budget	Opportunity cost is invisible

These symptoms share a feature: the information needed to stop the project usually exists inside the organization. What is missing is a decision process that surfaces it and acts on it.

THE METHOD

The Behavioral Stage-Gate

It adapts a familiar tool, stage-gate funding, by designing out the biases that normally defeat it.

Conventional gates often fail because criteria are vague, sponsors control the review and stopping is treated as a failure. The six design rules that follow address each of these.



Funding in tranches, with pre-agreed criteria deciding at each gate.

THE SIX DESIGN RULES

PRE-COMMIT
Set kill criteria first

PRE-MORTEM
Imagine failure in advance

TRANCHES
Fund in evidence-based stages

INDEPENDENCE
Separate deciders from sponsors

REFRAME
Stopping is reallocation

OUTSIDE VIEW
Anchor on base rates

THE DESIGN RULES

PRE-COMMIT

Pre-commit to kill criteria

Decide what evidence would end the project before anyone is emotionally or financially invested.

IN PRACTICE · At approval, require a one-page statement of the specific, measurable conditions under which the project will be stopped or re-scoped. Include tripwires such as spend exceeding a set share of the tranche without the milestone being met, a forecast benefit falling below break-even, or a critical assumption being disproved. Record the criteria, name an owner and treat them as a contract with the organization. Pre-commitment works because it moves the hard decision to a calm moment, much as automatic savings plans do.

PRE-MORTEM

Run a pre-mortem before funding

Imagining failure in advance makes it safe to voice doubts.

IN PRACTICE · Before approval, ask the team and independent reviewers to assume that the project has failed spectacularly a year from now and to write down why. Klein (2007) popularized this "premortem," drawing on research by Mitchell, Russo and Pennington (1989), who found that imagining an event as having already happened increased the number of reasons generated for it by roughly 30%. Treat that figure with care: the study measured the quantity of reasons, not their accuracy, and the broader evidence base for premortems is thin. Use the technique for what it reliably does, which is to surface concerns that would otherwise stay unsaid, and feed the top risks into the kill criteria. Repeat a short version at each gate.

TRANCHES

Fund in evidence-based tranches

Each tranche should buy information that reduces the biggest uncertainty, and unspent money should return to the pool.

IN PRACTICE · Break large programs into stages, each with a defined question ("does the core technology work at scale?", "will customers pay?") and a capped budget. Release the next tranche only when the evidence meets the criteria. This treats the project like a series of options: you pay a small amount to learn whether the next stage is worth buying, rather than committing to the whole.

THE DESIGN RULES, CONTINUED

INDEPENDENCE

Separate decision-makers from sponsors

People who own a project's history should not be the sole judges of its future.

IN PRACTICE · Convene an independent gate panel, drawing on finance, peer business units and a technical reviewer. Sponsors present the evidence but do not vote. Rotate panel membership and require that a project continued past a gate be justified against the question, **"Knowing what we know now, would we fund this from scratch today?"** This rule counters the escalation effect that Staw and others documented in decision-makers who are responsible for earlier choices.

REFRAME

Reframe termination as reallocation and learning

People will not propose stopping things if stopping ends careers.

IN PRACTICE · Call early terminations "capital reallocations," publish a short lessons-learned note for each, and celebrate decisions to stop at the earliest gate. Redeploy team members first, so that stopping is not equated with job loss. Track and recognize leaders who stop projects based on evidence. This rule depends on psychological safety (see The ROI of Psychological Safety), because a team that fears blame will conceal the signals that tell you to stop.

OUTSIDE VIEW

Use the outside view

Anchor forecasts on how similar projects have actually performed.

IN PRACTICE · Require every proposal to state a reference class (for example, "ERP implementations of this size") and the typical overrun and benefit shortfall in that class. Ask for the forecast to be justified against that base rate. Run a stress test: could the company absorb a 200% overrun if this became a black swan, as in Flyvbjerg and Budzier's data? If not, the project is too big to fund in one piece.

TOOL

A gate review template

Every criterion is set at approval, before anyone is invested. At each gate the panel compares the evidence with the threshold and makes one of three calls.

GATE CRITERION	EVIDENCE REQUIRED	THRESHOLD (SET AT APPROVAL)	DECISION RULE
Technical feasibility	Working prototype tested at the target scale	Meets the performance target	**Pass · stop · re-scope**
Customer or user value	Pilot results or signed commitments	Reaches the adoption or revenue target	**Pass · stop · re-scope**
Cost to complete	Independent estimate against the reference class	Within the agreed range of the tranche	**Pass · stop · re-scope**
Benefit case	Updated forecast with explicit assumptions	At or above break-even	**Pass · stop · re-scope**
Strategic fit	Panel assessment of alternatives	Still among the top uses of the capital	**Pass · stop · re-scope**

THE QUESTION AT EVERY GATE

"Knowing what we know now, would we fund this from scratch today?"

CASE ILLUSTRATION

Calder Insurance Group

A composite illustration, not a report on a named organization. All figures are illustrative.

A hypothetical company with a portfolio of 38 strategic initiatives and about $120 million in annual spend. Over three years it had cancelled none of them.

0 of 38	9	11 months
initiatives cancelled in three years	overran their budgets by more than 100%	median time from first red status to any decision

WHAT THE CFO AND PORTFOLIO OFFICE INTRODUCED

Criteria and pre-mortems	Every project at or above a size threshold had to document kill criteria and complete a pre-mortem before its next funding release.
Tranches	Large projects were re-baselined into three stages, with money released only against evidence.
Independent panel	A quarterly panel of finance, peer-unit and technical reviewers replaced the sponsor-led steering committees. Sponsors presented but did not vote.
Reallocation story	The CEO announced that stopping a project on evidence would be treated as a good decision, and team members of stopped projects were redeployed before any hiring.
Outside view	Each proposal had to quote the typical overrun for its reference class.

CALDER INSURANCE GROUP, CONTINUED

THREE LESSONS

- ✓ Some sponsors initially wrote vague kill criteria ("satisfactory customer response"), and the panel returned them until each had a number and a date.
- ✓ Teams began to sandbag early milestones to make gates easier to pass, which the panel countered by reviewing how milestones were set.
- ✓ The first project to be stopped was publicly recognized for the quality of the decision, which made the second one much easier.

ILLUSTRATIVE RESULTS AFTER EIGHTEEN MONTHS

METRIC	BEFORE	AFTER
Projects terminated at gates (of 38)	0	**7** (about 18%)
Median months from first red status to decision	11	**3**
Capital redeployed to higher-ranked initiatives	None	**$14 million** (about 12% of annual spend)
Projects with kill criteria set before funding	5 of 38	**38 of 38**
Projects over budget by more than 100%	9	**3** (all in the earliest stages)

The company stressed two caveats. A post-hoc review found that one of the seven terminated projects might have succeeded with more time, so the panel added a rule to document any real-option value before a stop. And it noted that eighteen months is too early to judge the long-term effect on portfolio returns.

MEASUREMENT

What to track

Track the speed and quality of stopping decisions, not just the count. A target number of cancellations would itself be a distortion, so include guardrails that detect both too few and too many stops. Baselines and targets are illustrative; replace them with your own measurements.

KPI	DEFINITION	ILLUSTRATIVE BASELINE	TARGET (18 MONTHS)
Time to decision	Median months from first red signal to a continue/stop decision	9–14 months	2–4 months
Kill criteria coverage	Share of funded projects with written criteria before funding	Below 20%	100%
Pre-mortem coverage	Large projects with a pre-mortem before funding	Below 10%	100%
Projects stopped at gates	Share of the portfolio stopped on evidence per year	0–3%	10–20% (an outcome, not a quota)
Capital redeployed	Spend released from stopped projects and reassigned	Not tracked	8–12% of annual spend
Overrun rate	Share of projects over budget by more than 100%	15–25%	Below 8%
Forecast accuracy	Gap between forecast and reference-class outcomes	Not tracked	Narrowing each year
False-kill rate Guardrail	Stopped projects that a later review judges viable	Not tracked	Reviewed annually; below 15%
Reporting candor Guardrail	Team-level safety score on raising bad news	Measure at start	Improving

A SIMPLE BUSINESS CASE

Suppose a $100 million portfolio includes projects that should have been stopped a year earlier, accounting for 10% of annual spend, or $10 million. If better gates recover half of that for higher-return work, the organization releases roughly **$5 million a year**. Add the avoided cost of the rare catastrophic overrun, which is much harder to estimate but potentially far larger. State all assumptions explicitly and compare them with the cost of running the panel and the pre-mortems, which is typically small.

GUARDRAILS

Where the intervention goes wrong

Killing too much. Long-horizon bets and genuine option value can look bad at early gates. Document the learning value of each stage and distinguish a project that is failing from one that is simply early.

Quotas. A target number of cancellations invites gaming and unfair stops. Treat the stopping rate as an outcome to monitor, not a goal.

Criteria gaming. Sponsors can set low bars, vague criteria or sandbagged milestones. Require numeric thresholds, review how milestones are set and have the panel sample earlier decisions.

A panel that turns political. If panel members have their own agendas or are rewarded for defending peers, the gate becomes another negotiation. Rotate members, publish decisions with reasons and include an outside voice.

Punishing the messenger. If stopping is treated as the sponsor's failure, bad news will be hidden. Pair this program with the psychological-safety practices in The ROI of Psychological Safety.

Hindsight bias. After a stop, people will say it was obviously right or wrong. Record the reasoning and the evidence at the time of each decision so it can be fairly evaluated later.

Treating people as costs. Teams on stopped projects are people with careers, not line items. Be honest about what happens to them and redeploy before cutting.

LEGAL AND COMPLIANCE NOTE

Stopping a project can trigger contractual, accounting and employment consequences. Contracts with vendors and partners may include minimum commitments, notice periods or termination fees. Capitalized development costs may need to be tested for impairment when a project is abandoned under IFRS or US GAAP, so involve finance and the auditors early. Where stopping leads to role changes, follow applicable employment-law consultation and notice requirements. Public companies should consider disclosure obligations for material write-offs. This paper is not legal or accounting advice.

THE PLAN

The first 90 days

DAYS 1–30

Diagnose

- List the top 15 to 20 initiatives by spend and, for each, record the original business case, the current forecast and the number of re-baselines.
- Identify the three to five initiatives with the weakest evidence and the longest time since their last honest review.
- Measure the baseline KPIs: time to decision, criteria coverage and overrun rate.
- Secure CEO and CFO backing for the principle that stopping on evidence is a good decision.

DAYS 31–60

Design and pilot

- Draft the kill-criteria template, the gate review template and the panel charter, with named members from outside the sponsoring units.
- Run pre-mortems on the three to five weakest initiatives and on one new proposal.
- Hold the first panel review and apply the "would we fund this today?" question to each initiative.

DAYS 61–90

Decide and embed

- Make and publish the first continue, re-scope or stop decisions with their reasoning.
- Redeploy the capital and the people from any stopped initiative and share the lessons learned.
- Require kill criteria and a pre-mortem for every new funding request above the size threshold.
- Set a quarterly cadence for the panel and report the KPIs to the board.

CALL TO ACTION

Pick the one initiative everyone privately doubts and ask a single question at the next review: if we were deciding today, with no history, would we fund it? Write down the answer and the evidence behind it. That one page is the start of a stage-gate system and often the most valuable document the program produces.

CONTINUE READING

Connections to the series

The ROI of Psychological Safety	Pre-mortems and honest status reporting only work when people feel safe delivering bad news.
The Nudge Enterprise	Make "stop" the default by giving projects sunset dates that expire unless actively renewed.
Cognitive Load and the C-Suite	Well-designed gate agendas and decision briefs reduce the load on executives making these calls.
The Halo and the Horn	Evaluating project leaders on the quality of their decisions, not on the outcome of a single project, avoids penalizing honest stops.
Atomic Habits at Scale	A regular kill log and quarterly gate review turn good decision hygiene into routine.

Brain Interacts

ABOUT BRAIN INTERACTS

Brain Interacts is a team analytics and behavioral analysis company. We build Corporate Brain and work with organisations on culture studies, root-cause analysis, and behavior-led operating design.

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